THE RISE AND PERSISTENCE OF ILLEGAL CROPS: EVIDENCE FROM A NAIVE POLICY ANNOUNCEMENT

Mounu Prem, Juan F. Vargas, and Daniel Mejía*

Abstract—Policies based on prohibition and repression to fight the war on drugs have largely failed in a variety of contexts. However, incentive-based policies may also fail and have unintended negative consequences if policymakers do not properly anticipate behavioral reactions. This is an particularly important concern in the case of policies announced prior to their implementation. In this paper, we show that a naive and untimely policy announcement generated an unprecedented escalation in cocaine production in Colombia, offsetting almost 20 years and billions of dollars of U.S.-backed efforts to stop drug production and cartel action.

I. Introduction

POLICIES that target beneficiaries based on observable characteristics may have unintended negative effects if they fail to anticipate people's equilibrium behavioral response. This is a particular concern for policies that are announced prior to their implementation. In anticipation of those policies, people may behave in ways that undermine their intended objective. This paper studies this problem in a setting where such badly designed policies can have enormous social costs and large externalities: the war on drugs.

Despite more than $10 billion spent by the American government on counter-narcotics efforts under Plan Colombia since 2000 (Government Accountability Office, 2018), by 2017 the amount of land cultivated with coca—cocaine's main precursor—exceeded the amount prior to the launch of Plan Colombia. This surge in coca growing occurred in just four years after the all-time lowest levels were reached in 2013. What explains Colombia's return to square one in the war on drugs after 20 years of costly drug enforcement efforts? We find that the recent and unprecedented expansion of coca was the result of an anticipation effect generated by the announcement of a future policy providing material incentives to replace growing coca with legal crops.

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In October 2012, the government of Colombia and the Revolutionary Armed Forces of Colombia (FARC from the Spanish acronym) started formal peace negotiations in Havana, Cuba. One of the most important milestones of the peace process was a partial agreement on how to tackle drug production and trafficking. In May 2014, the negotiating parties held a joint press conference in which they announced that, after reaching a peace agreement, the government was going to provide material incentives to switch from growing coca to other crops.

We estimate the effect of this announcement on coca growing by exploiting the time variation given by the press release, along with two sources of cross-sectional variation. First, we rely on an ecological time-invariant and continuous coca suitability measure to distinguish municipalities based on their differential cost advantages of growing coca and thus their ability to respond to the anticipated benefits. Second, using exactly the same information that coca growers had at the time of the May 2014 announcement, we predict which areas would be targeted by the crop substitution program. Specifically, the announcement stated that incentives were to be targeted at poor areas with a history of growing coca.[1] We thus complement our measure of coca suitability with one of predicted material benefits.

We exploit both the temporal and the geographical sources of variation in a *difference-in-differences* empirical framework, which also controls for municipality fixed effects, department-year fixed effects, and a large set of predetermined municipal characteristics interacted with the postannouncement time dummy. We find a large differential increase in the area cultivated with coca in coca-suitable municipalities as well as in places with a higher probability of receiving material incentives for crop substitution. Using our most demanding specification we find that a one-standard-deviation increase in coca suitability increases coca growing by almost one-quarter of a standard deviation after the 2014 announcement, a magnitude that is equivalent to the mean of coca cultivation during our sample period. Moreover, if we increase by 50% the estimated probability that a municipality will receive the incentives for crop substitution relative to the mean (from 5% to 7.5%), then the area of coca crops would increase by 0.57 standard deviations. These findings are robust to exploiting grid-level variation in coca plantations within municipalities, which allows us to control for municipality-year fixed effects.

We rule out a number of alternative potential explanations. First, we show that coca growing after the policy announcement occurs both in grids adjacent to old (preannouncement)

[1] A translated relevant transcript of the announcement can be found in appendix A.1.

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crops as well in new detached ones. The latter suggest that the coca expansion is not driven by private information available to traditional coca growers, but by the public incentives generated by the announcement. We also rule out that contemporaneous changes in policy initiatives against the war on drugs are driving the results. These include the 2015 ban of aerial spraying of coca fields, the confiscation of coca paste and cocaine shipments, and dismantling of clandestine coca-processing laboratories. Moreover, we show that the behavior of rural farmers is not explained by the suitability of land to grow key legal crops promoted by the government to replace coca or by other profitable economic activities such as gold mining.[2] Finally we document the robustness of our findings to the differential roll-out of antipoverty and social programs and rule out the potential endogeneity of coca suitability to preannouncement deforestation trends.

The differential surge of coca growing in coca-suitable municipalities or in places with a higher probability of benefiting from coca substitution incentives increases over time and during our entire postannouncement period (ending in 2018). This is important because the program was actually implemented in 2017, after the final peace agreement was signed. Thus, above and beyond the anticipation effect generated by the announcement, the material incentives actually provided by the program itself created further expectations among nontargeted farmers, encouraging them to grow coca.

Although our paper is not about how incentive-based policies influence the development of criminal organizations, our findings do suggest, however, that this policy mistake may have indirectly affected how criminal organizations operate in Colombia. It is important to emphasize that the channel through which this happened is not the criminalization of growers, since in most drug-producing countries—including Colombia—growers of coca and poppies cannot be imprisoned (Farrell, 1998). Instead, drug traffickers in charge of transforming coca leaves into cocaine and shipping it abroad will likely be attracted to the large increase in the supply of coca leaves. In fact, we show that violence inflicted by other armed groups—which did not take part of the peace process—increased substantially after the press conference, and it did so specifically in the areas that reacted to the announcement by increasing coca growing. Together with the inadequate implementation of the crop substitution program, this explains the persistence of coca growing even after the program was launched.

These results are important for the discussion on the most effective ways to fight drug production and trafficking, a question that has been at the core of the policy debate since U.S. President Richard Nixon declared the war on drugs in the 1960s. For decades, the war on drugs has by and large been carried out through prohibition, punitive means, and enforcement efforts. It has largely failed (Baum, 1996; Gray, 2013). Among other unintended costs, prohibition encourages violence (Miron & Zwiebel, 1995). This occurs, for instance, because business disputes in illegal markets cannot be resolved through the legal system. Miron (1999, 2001) provide empirical evidence on the relationship between prohibition and violent crime, respectively, for the United States and for a cross section of countries. Moreover, in Mexico, Afghanistan, and Colombia different drug enforcement policies have resulted in more violence, respectively by creating violent power disputes (Calderón, Robles, Díaz-Cayeros, & Magaloni, 2015; Dell, 2015), by increasing illegal profits (Clemens, 2013), and by generating retaliation from violent actors affected by the policy (Abadie et al., 2013). Lessing (2017) argues that antidrug repressive policies in Brazil, Colombia, and Mexico generated incentives for drug cartels to fight back, increasing violence and corruption instead of consolidating the rule of law.[3]

The war on drugs has relied on incentive-based policies such as the one studied in this paper to a much lesser extent. But this does not imply that a shift of policy instruments (from prohibition to incentives) would unambiguously make governments better equipped to curb drug production and trafficking. As mentioned, incentives and regulation can also go wrong if policymakers fail to anticipate the equilibrium behavioral response beyond their targeted goal. Moreover, targeted benefits can encourage noneligible people to select into the group of beneficiaries, often by changing their behavior and creating distortions. For example, when government subsidies are targeted using proxy-means tests to identify beneficiaries, this may discourage people from getting out of poverty.

Our findings also have implications for the way in which policies are discussed. For example, although making parliamentary debates public may promote accountability and provide key information for voters, it has the drawback of potentially creating announcement effects that can make policies ineffective or even have unintended harmful consequences.[4]

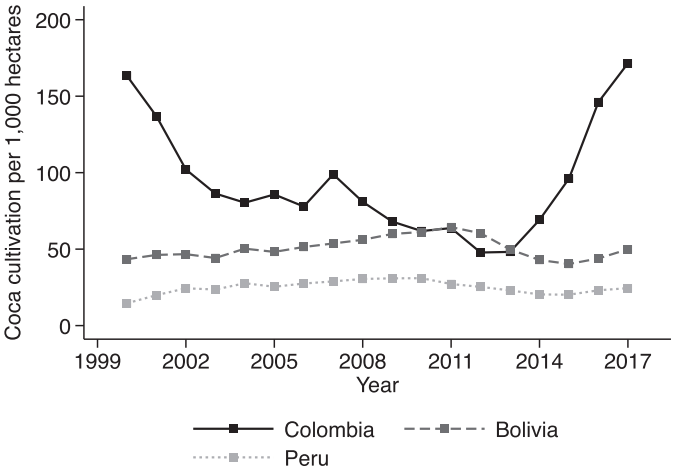
We also contribute to the literature that evaluates the effectiveness of antidrug efforts in reducing illegal drug production. Most of these studies have focused on the direct and collateral effects of aerial spraying efforts aimed at reducing coca cultivation. This literature finds that drug eradication efforts are ineffective in reducing the size of coca plantations (Abadie et al., 2013; Reyes, 2014; Mejía, Restrepo, & Rozo, 2015), increase violence (Mejía & Restrepo, 2015; Angrist & Kugler, 2008; Díaz & Sánchez, 2004; Abadie et al., 2013), have negative health effects (Camacho & Mejía, 2017), and create environmental degradation (Dávalos, Sanchez & Armenteras, 2016).

[2]Potentially, even if we account for the main potential substitutes for coca growing, a risk will also be that concurrent changes in the price of other commodities may create some minor bias.

[3]The relationship between crack-down-type policies and violence also applies to integrated markets across different countries. For example, Castillo, Mejía, and Restrepo (2018) show that cocaine seizures in Colombia produced scarcity in downstream drug markets in Mexico and, by increasing prices, increased drug-related violence in Mexico.

[4]An illustrative example is the "Safer Neighborhoods Gun Buyback Act of 2019," which provides material incentives in exchange for firearms in the United States.

FIGURE 1.—EVOLUTION OF COCA CULTIVATION



This figure presents the evolution of coca cultivation per 1,000 hectares in Colombia for the period 2000–2017.

We also contribute to the literature that evaluates the effectiveness of alternative development programs. Farrell (1998) reviews the effects of such programs in eleven opium- and coca-producing countries (including Colombia) and concludes that most of them are ineffective. Multiple reasons can be identified for the failure of such programs, including the lack of long-term planning and continuous redesign of policies with every new administration (Vargas, 2005), the inadequate market appraisal of viable alternative crops and the backfiring of conditional credits (Lupu, 2004), and the lack of engagement with strengthening local democratic institutions (Farthing & Kohl, 2005).[5] Beyond the implementation of these policies, we emphasize the importance of the way they are announced.

II. Context

Coca crops are the main input into the production of cocaine, and Colombia has been the world's top cocaine producer for the last three decades. In September 1999, the governments of Colombia and the United States launched a joint strategy called Plan Colombia, aimed at combating illegal drug production and recovering the control of large areas of the country that were dominated by illegal guerrilla and paramilitary armed groups. The main source of funding for these groups was producing cocaine and shipping it to the United States and Europe. According to the United Nations Office for Drugs and Crime (UNODC), at that point Colombia 163,000 hectares of coca crops (see figure 1).[6] Albeit with some ebb and flow, coca cultivation declined substantially in the following years, reaching 48,000 hectares in 2013. However, from 2014 to 2017 coca crops increased exponentially, reaching a historical peak (171,000 hectares).[7]

Alongside with the evolution of coca production in Colombia, figure 1 shows the evolution of coca production in Peru and Bolivia, the only other coca-producing countries. Importantly, neither of them had anything similar to the extraordinary increase in coca growing that occurred in Colombia since 2014 (the lines are in fact pretty much flat through the sample period). This provides suggestive evidence that an abrupt change in the international demand for cocaine is unlikely to have taken place. To complement this evidence, we collected data on the annual prices paid at various cocaine production stages in Colombia. Their evolution is plotted in figure A5 of the appendix. The figure suggests that although the price of coca leaves seems to be cyclical both before and after the announcement, with no apparent postannouncement differential pattern, the prices of coca paste and coca hydrochloride (cocaine for consumption) are rather stable over the sample period. Again, this suggests that international demand is an unlikely explanation for our findings.[8]

What explains then the unprecedented surge in coca growing since 2014? We argue that this trend was triggered by the anticipation effect generated by the announcement, during the peace process, that coca growers were to receive future material benefits should they switch their crops. In May 2014, the Colombian government and FARC held a joint press conference where they announced the eventual creation of a crop substitution program. Critically, the announcement included the wording "material welfare conditions for the people affected by illicit crops." According to recent journalistic accounts of how the secret peace dialogues developed in Havana, it was only *after* the press conference was broadcast that the government team realized that the announcement could have created perverse incentives (Bermúdez Liévano, 2018). In any case, when the Comprehensive National Program for the Substitution of Crops for Illicit Use (PNIS for its name in Spanish) was launched in 2017, coca growers did end up receiving direct cash transfers from the government.[9]

[5]Other authors point to positive effects of alternative development programs, emphasizing the key role of social development, poverty reduction, and community-based activities to create social capital (Cohen, 2009; Dispanadda Diskul et al., 2019), as well as that of price premiums inherent to organic and fair-trade certified products (Garcia-Yi, 2014).

[6]One hectare is roughly 2.5 acres.

[7]Potential cocaine production also reached an all-time record (1,058 metric tons).

[8]The stability of prices implies that the surge in coca production that took place in Colombia between 2013 and 2017 did not cause a substantial decrease in the retail price of cocaine. This is because drug traffickers alter the purity of cocaine in response to changes in the supply of coca leaves in order to keep the supply of retail cocaine constant and avoid price drops (Londoño, 2020). This is done by using more or fewer chemical precursors in the transformation of coca paste into cocaine.

[9]The qualitative evidence also supports the view that rural communities got the message that growing coca would yield future government benefits. According to the nonpartisan think tank Fundación Ideas para la Paz, several community-level organizations such as Juntas de Acción Comunal—present throughout the entire country—de facto assumed the role of keeping the communities on top of what was going on in the peace negotiation and eventually of the content of the agreement. These organizations spread the idea that families that had coca crops in their land would receive material benefits from the government in exchange for eradicating these crops (see, e.g., Garzón & Wilches, 2015; Garzón, 2019; Garzón, Gelvez, & Bernal, 2019).

Importantly, as discussed in appendix A.2, the timing of the announcement was not confounded by any other milestone of the peace process during the few months preceding or following it. One important potential confounding factor is, however, the ban of aerial spraying that the National Anti-Drug Council enacted in May 2015, based on the extensive evidence that the use of glyphosate in the aerial spraying of illicit crops was associated with the incidence of cancer and other diseases (International Agency for Research on Cancer, 2017; Camacho & Mejía, 2017; Dias, Rocha, & Soares, 2019). This decision was opposed by several political groups, including the main opposition party in Colombia as well as the U.S. government. Critics argued that the suspension of aerial spraying caused the surge in coca cultivation. This led to diplomatic pressure from the United States to resume aerial eradication. We test this (and other) alternative explanations of the increase of coca crops and do not find any empirical support for them.

III. Data

A. *Area Cultivated with Coca and Suitability*

Our main outcome variable is the share of hectares of coca crops relative to the size of the municipality in thousands of hectares. This is a satellite-based measure estimated annually since 1999 by the Integrated Monitoring System of Illicit Crops (SIMCI from its name in Spanish) of the UNODC. We describe in detail SIMCI's estimation procedure in appendix A.3. Table A1 shows that, during our sample period, the average municipality had 0.63 hectares of coca per thousand hectares of municipal land (standard deviation 3.78).

In addition, we exploit cross-sectional variation across municipalities on the average suitability of the land for growing coca. This variable was constructed by Mejía and Restrepo (2015), and it is based on various rounds of a nationally representative household survey of coca farmers conducted by SIMCI/UNODC between 2005 and 2010. Surveyed coca growers were randomly selected using the satellite estimates described above to identify their location. In total, 1,678 farmers were surveyed in 64 municipalities scattered around the country. The household survey provides self-reported data on coca crop yields, which is then combined by Mejía and Restrepo (2015) with exogenous municipal geographic and weather characteristics to estimate an index of each municipality's suitability for coca growing. These include the altitude of the municipality, a soil erosion index, a soil aptitude index based on soil nutrients, minerals, and the topography of the municipality and average rainfall levels. For this paper, we also created an original measure of coca suitability at the grid level that we describe in appendix A.4.

B. *Predicted Targeting of the Incentive Scheme*

Our second source of cross-sectional variation exploits the extent to which farmers could anticipate—at the time of the announcement—the likelihood that they would benefit from the substitution program. In the spirit of our coca suitability measure described above, we constructed a predicted probability that a given municipality will be targeted by the coca substitution program. To do this, we estimated a probit model using an indicator of the actual posterior targeting of PNIS as dependent variable.[10] Following the press release (see appendix A.1), we use a poverty index and the historical incidence of coca crops as regressors, both measured before the announcement. We find that both variables are positively associated with the eligibility and the implementation of PNIS (see table A2). Table A1 presents the summary statistics for the predicted probability of receiving material incentives, which has an average of 5%. Similarly, 5% of the country's municipalities ended up receiving the program in 2017.

C. *Conflict Data*

We examine violence as a potential mechanism of the dynamics of coca growing after the initial boost given by the anticipation behavior of coca growers following the announcement. To this end, we use a conflict data set originally compiled by (Restrepo, Spagat, & Vargas, 2004) and updated through 2018 by the Universidad del Rosario. This data set codes violent events recorded in the Noche y Niebla ("Night and Fog") reports from the NGO Centro de Investigación y Educación Popular (CINEP) of the Society of Jesus in Colombia, which provides a detailed description of the violent event, its date of occurrence, the municipality in which it took place, the identity of the perpetrator, and the number of victims involved in the incident.[11]

D. *Coca Eradication*

The most important potential confounder that we examine is the ban on aerial eradication. To that end, we constructed municipal-level cumulative aggregates of the size of the area of illicit crops sprayed by Colombian authorities. The source of these data is the Anti-Narcotic Unit of the National Police.

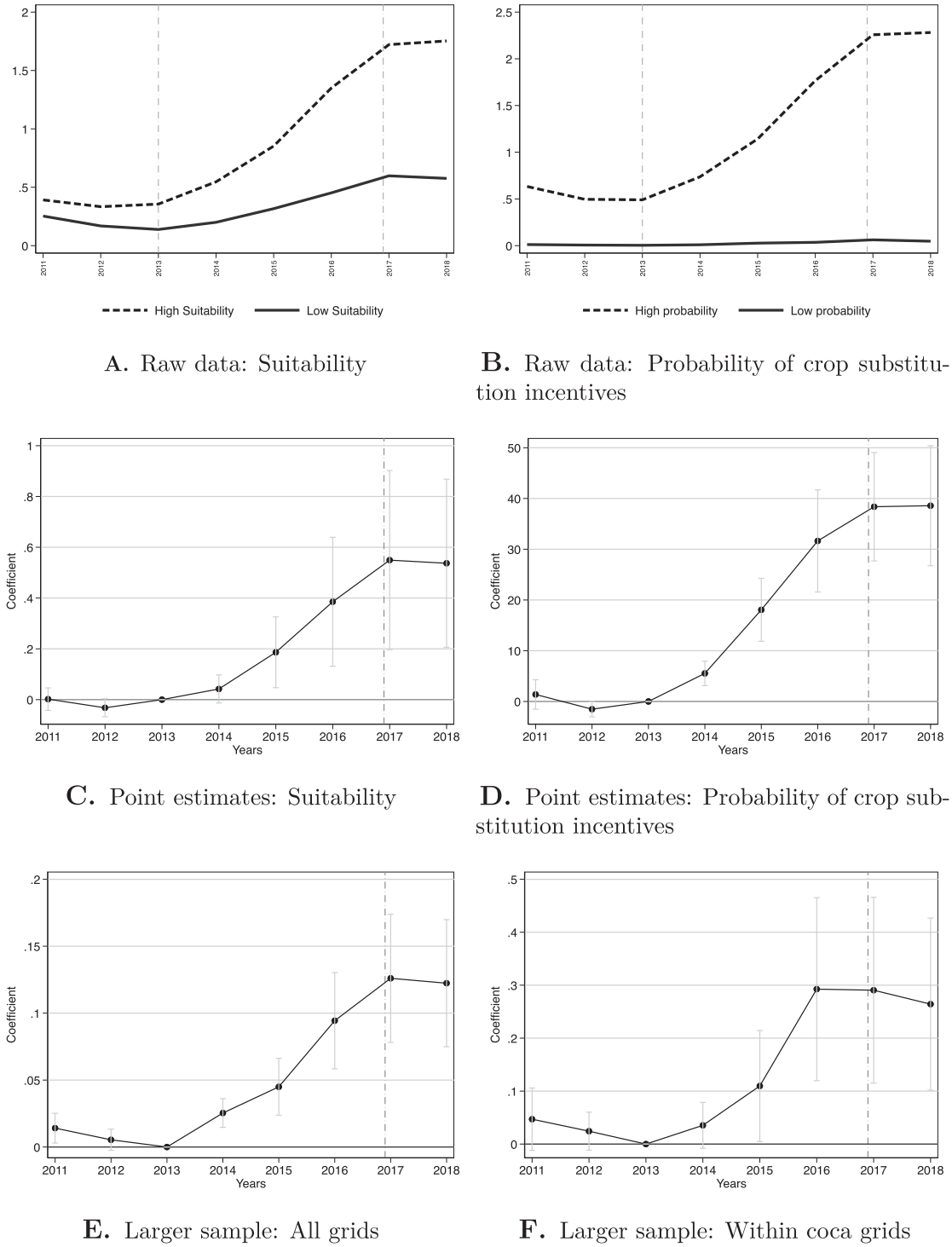
E. *Other Data*

Our preferred specification controls for a large set of predetermined municipal characteristics interacted with the postannouncement period dummy. These control for differential changes across municipalities and over time,

[10]For our baseline results we use the 77 municipalities listed by the government as eligible for PNIS in 2017 in a kind of intent-to-treat approach (see Alta Consejería Presidencial, 2017). All our results are, however, robust to using only the 56 municipalities that actually ended up receiving the program by the first half of 2019 (see UNODC, 2019).

[11]Noche y Niebla sources include "1. Press articles from more than 20 daily newspapers of both national and regional coverage. 2. Reports gathered directly by members of human rights NGOs and other organizations on the ground such as local public ombudsmen and, particularly, the clergy" (Restrepo et al., 2004, p. 404). Notably, since the Catholic Church is present in even the most remote areas of Colombia, we have extensive coverage of violent events across the entire country.

FIGURE 2.—RAW DATA DYNAMICS AND POINT ESTIMATES



A. Raw data: Suitability

B. Raw data: Probability of crop substitution incentives

C. Point estimates: Suitability

D. Point estimates: Probability of crop substitution incentives

E. Larger sample: All grids

F. Larger sample: Within coca grids

This figure presents the evolution of coca area for municipalities with high and low coca suitability and for municipalities with high and low probability of benefiting from crop substitution incentives. Panels A and B present the raw data. Panels C and D present the coefficients from our dynamic specification presented in equation (2). Panel C uses our continuous treatment of coca suitability, and panel D uses the probability of benefiting from crop substitution incentives. Panels E and F report the coefficients from the dynamic specification using predicted coca suitability at the grid level (as discussed in section III.A). Panel E includes all the grids in our main sample of municipalities, while panel F uses only grids that had coca at some point after 2011. For panels C to F, we present the point estimates of the regression and the confidence interval at the 95% level.

parametrized by variables that characterize the municipal size, the degree of rurality, poverty rates, and geographical segregation.

F. Descriptive Statistics

Panels A and B of figure 2 present the evolution of the share of coca cultivation for a discrete version of our two

sources of cross-sectional variation, namely, coca suitability and the predicted probability of benefiting from coca substitution incentives. In both cases we define "high" (highly suitable or with a high predicted probability) as the municipalities above the median of the empirical distribution of these measures. Not surprisingly, we can see that in both cases the places with high suitability and high predicted probability of receiving material incentives for crop substitution have more coca cultivation before the announcement. However, visual inspection suggests that despite this level differences the trends follow a similar trajectory.[12] One also sees, after the press release, a differential increase in coca cultivation in areas highly suitable or with a high predicted probability.

Figure 3 further explores the relationship between the increase in coca and our two measures of what places could have been most affected by the anticipation effect generated by the policy announcement. In blue, we present the change in average coca cultivation between 2014–2018 and 2011–2013. Darker blue shades mean a larger interperiod increase. In turn, the red dots present suitability (panel A) and predicted probability of receiving crop substitution incentives (panel B). Larger dots represent higher suitability and a higher predicted probability, respectively. The patterns in both maps suggest that the increase in coca cultivation was higher in places with lower costs of growing coca (because they have a high suitability) or with high potential benefits (because the predicted probability is high). In the rest of the paper, we explore this relationship more formally.

IV. Empirical Strategy

Our identification strategy exploits the timing of the preannouncement of future incentives to substitute away coca, as well as the cross-sectional variation provided by a measure of the cost advantage of growing coca (the coca suitability index) and by a measure of the expected benefits of the crop substitution program (the predicted probability of receiving material incentives for crop substitution). More formally, using the subindex m to denote municipalities, d to denote departments, and t to denote time, we estimate the following difference-in-differences model:[13]

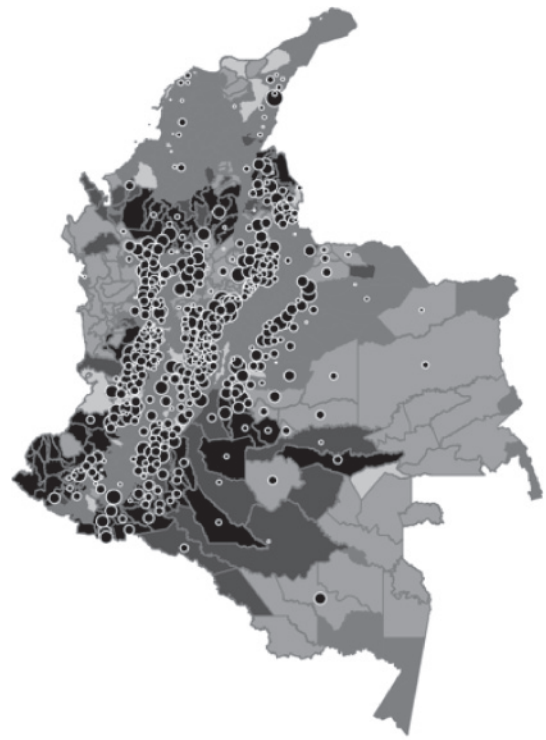
$$y_{mdt} = \alpha_m + \lambda_{dt} + \beta(Announcement_t \times T_m) + \sum_{c \in \mathbf{X_m}} \gamma'(c \times Announcement_t) + \varepsilon_{mdt}, \quad (1)$$

where y_{mdt} is measures the share of the municipality area with coca plantation per 1,000 hectares, T_m measures either the

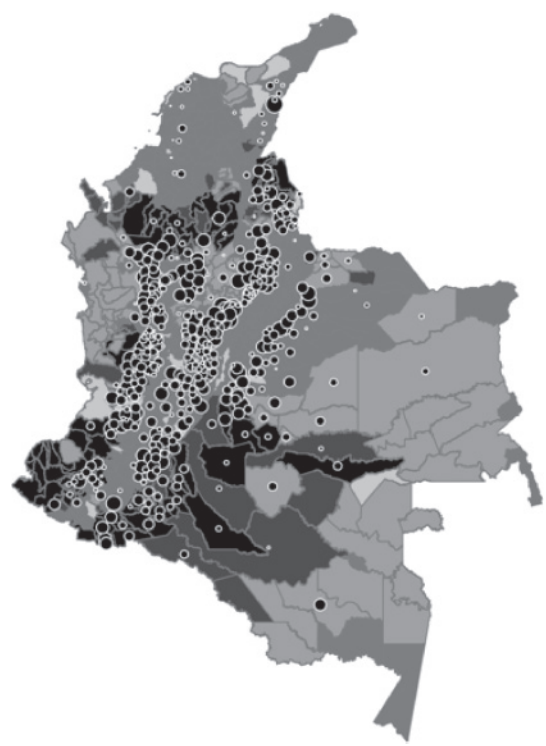
[12]Because this speaks to the main identification assumption of our methodology, we will provide a set of more formal tests below.

[13]Municipalities are equivalent to U.S. counties and departments to U.S. states. There are around 1,100 municipalities in Colombia distributed in 32 mainland departments plus the Caribbean island of San Andrés. Our sample excludes the latter as well as the capital city, Bogotá, which constitutes its own department. We thus end up with 31 departments, with the average department having 35 municipalities.

FIGURE 3.—CHANGE IN COCA PRODUCTION, COCA SUITABILITY, AND SUBSTITUTION PROBABILITY



A. Suitability



B. Probability of receiving substitution incentives

This figure presents the change in coca production after the announcement compared to before. Darker blues means a larger increase in coca production after the announcement. In panel A, the red dots represent the coca suitability of the soil; larger dots means more suitability. In panel B, the red dots represent the expected probability of benefiting from crop substitution incentives; larger dots mean higher probability.

coca suitability of municipality m or the expected probability of benefiting from coca substitution incentives in municipality m, $Announcement_t$ is a dummy that takes the value 1 after the press release, α_m are municipality fixed effects, and λ_{dt} are department-year fixed effects.[14] X_m is a vector of various municipality characteristics measured prior to the announcement that we interact with the time indicator that identifies the postannouncement period to flexibly control for differential changes parametrized by each one of the municipal attributes included in the vector. Finally, ε_{mdt} is the error term, which we cluster at the municipality level in the case of the treatment based on coca suitability.[15]

Our coefficient of interest, β, captures the average differential change of coca growing before and after the announcement in municipalities with high coca suitability or in municipalities with a high estimated probability of receiving material incentives for crop substitution relative to municipalities with low suitability or a low estimated probability. Moreover, our estimation takes into account any municipality characteristic that does not vary over time, as well as any differential trends in coca cultivation that are different across departments. We also estimate an augmented version of equation (1) to study heterogeneous effects according to key municipal-level characteristics and shed some light regarding the underlying mechanisms of the main effects of interest.[16]

A. *Identifying Assumption*

The main assumption behind our difference-in-differences model is that, in the absence of the announcement, the area cultivated with coca in municipalities with high coca suitability or with a high estimated probability of benefiting from coca crop substitution incentives would have followed a similar trajectory to coca growing in municipalities with low coca suitability or a low estimated probability. The validity of this "parallel trends" assumption can be partially assessed by estimating the following nonparametric regression:

$$y_{mdt} = \alpha_m + \lambda_{dt} + \sum_{j \in J} \beta_j (T_m \times \delta_j) + \epsilon_{mdt}, \tag{2}$$

where J includes all years in our sample except from 2013, which is the year before the announcement. Therefore, the parameters β_j can be interpreted as the differential coca production in municipalities with high coca suitability or high predicted probability relative to municipalities with low suitability or probability in year j relative to the year prior to that of the policy announcement.

As shown by Goldsmith-Pinkham et al. (2020), identification in Bartik-type instruments could come from differential exposure to a common shock. This critique also applies to difference-in-differences designs such as ours. This may be problematic, however, if the differential exposure to the shock is correlated to other characteristics that predict future outcomes. As proposed by the authors, showing that the differential exposure is not correlated with differential pretrends helps alleviate this concern. This is precisely the purpose of estimating equation (2).[17] The authors propose a second solution, namely, to control for characteristics correlated with the exposure to the shock. To address this concern, we follow Belloni et al. (2014) and perform a machine learning algorithm that selects the controls that are better at predicting our measures of exposure to the announcement, as well as the evolution of coca crops. We control for these, interacted with the postannouncement time dummy. Our results are robust to the inclusion of these controls (see appendix A.6).

V. Results

A. *Main Findings*

Table 1 reports the coefficients of interest obtained from estimating equation (1). Columns 1 to 3 present the differential effect of the announcement on coca growing for municipalities with high coca suitability. Columns 4 to 6 do the same for municipalities with a high predicted probability of receiving material incentives for crop substitution. Columns 1 and 4 include municipality and year fixed effects, and columns 2 and 5 replace the latter for department×year fixed effects to capture any aggregate yearly shock at the department level. Finally, columns 3 and 6 are equivalent to 2 and 5 but also control for differential changes in coca cultivation after the cease fire due to several pre-cease-fire municipality characteristics.[18] The standard errors in parentheses are clustered at the municipality level in columns 1 to 3 and bootstrapped in columns 4 to 6. For robustness, in square brackets we report p values that take into account the potential cross-sectional dependence in the error term Conley (1999, 2016).[19]

Focusing on coca suitability as a measure of the relative cost advantage for coca cultivation, we find that, following the press release, a municipality with a coca suitability index one standard deviation higher than the average municipality experienced an increase in the share of the municipal area cultivated with coca of 0.30 hectares per 1,000 hectares (column 1). This increase is statistically significant and

[14] These set of fixed effects control respectively for any observed or unobserved municipal-level time invariant heterogeneity, and for any time shocks that simultaneously affect all the municipalities of the same department.

[15] When the cross-sectional variation is given by the predicted probability of benefiting from substitution incentives, we wild bootstrap the standard errors clustered at the municipality level given that this measure comes from the predicted values of a previously estimated regression model.

[16] We discuss the estimating equation in appendix A.5.

[17] In addition, for robustness we exploit variation in coca suitability within municipalities, which addresses the concern that municipality-level characteristics that correlate with municipality coca suitability drive our results.

[18] This set of municipality characteristics include the logarithm of population, the share of rural population, a poverty index, and the distance to the department capital.

[19] We also implement the standard errors suggested by Adao et al. (2019) that allow for any correlation structure of the regression residuals across geographical regions. For all columns we find the p values smaller than 1% (not reported).

TABLE 1.—COCA PRODUCTION, COCA SUITABILITY, SUBSTITUTION PROBABILITY, AND POLICY ANNOUNCEMENT

	Dependent variable: Share of coca cultivation over 1,000 hectares					
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment:		Coca Suitability			Probability of Substitution Incentives	
Treatment × Announcement	0.30**	0.35***	0.28**	26.99***	26.48***	27.38***
	(0.12)	(0.11)	(0.11)	(4.67)	(4.70)	(5.03)
	[0.001]	[0.001]	[0.006]	[0.000]	[0.000]	[0.000]
Observations	8,736	8,736	8,736	8,736	8,736	8,736
R-squared	0.683	0.736	0.749	0.366	0.438	0.441
Municipalities	1,092	1,092	1,092	1,092	1,092	1,092
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Department-Year FE	No	Yes	Yes	No	Yes	Yes
Controls	No	No	Yes	No	No	Yes
Mean DV	0.274	0.274	0.274	0.274	0.274	0.274
SD DV	1.240	1.240	1.240	1.240	1.240	1.240

This table presents the results from the main specification in equation (1). *Suitability* is an index of coca suitability constructed by Mejía and Restrepo (2015). *Probability of substitution incentives* is a predicted probability of benefiting from crop substitution incentives estimated in table A2. *Announcement* is a dummy that takes the value 1 for the period after 2013. Columns 3 and 6 add predetermined municipal controls interacted with the cease-fire dummy. These controls include logarithm of the population in 2010, share of rural population, poverty index, and distance to the department capital. Columns 1 to 3 present robust standard errors clustered at the municipality level in parentheses, while columns 4 to 6 present bootstrap standard errors. In square brackets, we present the *p* values for standard errors control for spatial and first-order time correlation (see Conley, 1999; Conley, 2016). We allow spatial correlation to extend to up to 279 km from each municipality's centroid to ensure that each municipality has at least one neighbor. * is significant at the 10% level, ** is significant at the 5% level, and *** is significant at the 1% level.

economically large. It represents 24% of a standard deviation and 109% of the sample mean of coca growing during the preannouncement period. Results are of similar magnitude and significance once we add department-year fixed effects (column 2) and control for differential changes in the outcome after the policy announcement parametrized by preannouncement characteristic (column 3). The results are also robust to accounting for spatial correlation in the error term.

Turning to the predicted probability of benefiting from coca substitution incentives as a proxy of the expected benefits of growing coca, we find similar results in terms of economic importance. In particular, following the press release, a municipality with a predicted probability 10% higher that the mean (which is equal to 5%) experienced an increase in the share of the municipal area cultivated with coca of 0.14 hectares per 1,000 hectares (column 4). Again, this increase is statistically significant and economically large. It represents 11% of a standard deviation and 49% of the sample mean of coca growing during the preannouncement period. As in the case of coca suitability, the results are of similar magnitude and significance once we add department-year fixed effects (column 5), control for differential changes in the outcome after the policy announcement parametrized by preannouncement characteristic (column 6), or account for potential spatial correlation in the error term.[20]

These results are largely robust to exploiting variation in coca suitability within municipalities. Table 2 shows estimates of a more demanding version of equation (1) that exploits our original machine learning estimates of coca suitability at the grid level (see appendix A.4 for details on its construction) and includes municipality × time fixed effects.[21] Column 1 replicates the municipality-level results of table 1, using a suitability measure that, in comparison with the grid-level suitability estimate, is computed using only weather-related and altitude variables. We do this because grid-level satellite-based data do not include soil quality indicators.[22] If anything, with this alternative measure, the municipality-level results are somewhat larger in magnitude relative to the baseline estimate in table 1, column 2.

TABLE 2.—WEATHER- AND ALTITUDE-BASED COCA SUITABILITY AND WITHIN-MUNICIPALITY ANALYSIS

	(1) Weather and altitude coca suitability	(2) All grids	(3) Within coca grids
Suitability × Announcement	0.429***	0.076***	0.175***
	(0.12)	(0.018)	(0.071)
Observations	8,736	8,867,184	898,152
R-squared	0.197	0.664	0.634
Municipality FE	Yes	No	No
Department-Year FE	Yes	No	No
Grid FE	No	Yes	Yes
Municipality-Year FE	No	Yes	Yes
Grids		1,108,398	112,269
Mean DV	0.274	0.055	0.469
SD DV	1.240	0.425	1.416

This table presents the results from the main specification in equation (1) for column 1 and at the grid-level for columns 2 and 3. The specification in columns 2 and 3 include grid-level fixed effects and municipality-year fixed effects. *Suitability* is defined as the predicted suitability at the municipality level in column 1 and at the grid level in columns 2 and 3, as discussed in section III.A. *Announcement* is a dummy that takes the value 1 for the period after 2013. Bootstrap standard errors are presented in parentheses. * is significant at the 10% level, ** is significant at the 5% level, and *** is significant at the 1% level.

[20]Recent literature has shown that linear regressions with two-way fixed effects can lead to positive coefficients, whereas the average treatment effects could be negative. We implement the computation of weights suggested by De Chaisemartin and d'Haultfoeuille (2020) with a discrete version of our treatment based on values above the median and find no negative weights, suggesting that this is not a major concern in our context.

[21]Specifically, we estimate

$$y_{gmt} = \alpha_g + \lambda_{mt} + \beta(Announcement_t \times Suit_{gm}) + \varepsilon_{gmt},$$

where g indexes the grid, and λ_{mt} are the municipality × time fixed effects. The rest of the notation is as in equation (1).

[22]An additional advantage of this alternative suitability measure is that, by excluding soil quality measures, it reduces concerns of a potential endogeneity of soil quality to previous coca cultivation, which, for instance, may create deforestation.

Columns 2 and 3 of table 2 report the grid-level results for all 1 km × 1 km grids in the country and for those that experienced coca growing at any point in time during our sample period, respectively. Focusing on the estimates that use all the grids of the country, we find that a one-standard-deviation increase in coca suitability increases coca growing after the announcement by 0.18 standard deviations or 138% of the preannouncement sample mean. These estimates are remarkably similar in magnitude to those reported in table 1 using municipality-level variation.[23]

In table A3 we asses the relative importance of our two sources of cross-sectional variation. We do so by including in the same regression model (equivalent to specification 1) the interaction of the postannouncement indicator with both variables simultaneously (column 1) and that plus the triple interaction (column 2). Our results are robust to the inclusion of both treatments simultaneously. This suggests that both treatments are independently relevant in this context and do not capture the same spatial variation.[24] The coefficient associated with the triple interaction, which can be interpreted as the differential change of coca growing following the announcement in places with both high suitability and expected to be targeted by the substitution program, is positive and significant (column 2). This implies that the treatments are complementary.

B. *Identifying Assumption*

Panels C and D of figure 2 report the coefficients that result from estimating the nonparametric version of equation (1) (that is, equation [2]). Panel C focuses on the effect of the press release on coca growing in municipalities with a higher coca suitability index. Meanwhile, panel D looks at the variation provided by the predicted probability of receiving material incentives for crop substitution. In both cases we see no differential trend in 2011 and 2012 with respect to 2013, the year prior to the policy announcement. This evidence provides support for the main identifying assumption of our empirical strategy.[25]

Importantly, in contrast to the estimated effects for the years prior to the announcement, starting in 2014 there is differential increase in the share of coca plantations in municipalities with high coca suitability or with a high estimated probability of receiving material incentives for crop substitution. Moreover, the effect grows overtime and slows down (but does not revert) only in 2018, the final year of our sample. Panels E and F of figure 2 show the equivalent graphical illustration of the dynamic specification for the case of the grid-level results.

[23]The economic magnitude of the estimates are somewhat smaller for the case of the subset of grids that recorded coca cultivation at any time.

[24]Indeed, the correlation between the two measures is rather small: 0.05.

[25]Figure A2 shows that the absence of pretrends is robust to adding a longer preannouncement period (starting in 2008). For completeness, columns 1 and 2 of table A4 in the appendix show a reestimation of the main specification using this longer sample period and find results that are largely consistent with those reported in table 1.

We also conduct a more parametric test for the existence of differential trends during the preannouncement period in the spirit of Muralidharan and Prakash (2017). In this test we interact a linear trend with our measures of having a low cost of growing coca and of the high potential benefits of doing so, and test for the significance of this coefficient prior to the announcement. Reassuringly, we find no evidence for differential trends before the cease ire for the probability of benefiting from the substitution program, and only a small and marginally significant coefficient for the suitability index (see table A5 columns 1 and 6). In addition, we conduct a placebo test, using the announcement of the partial agreement for political participation, which took place exactly one year prior to the announcement of the illicit crops agreement (see figure A1). This test is reported in columns 2 and 7 of table A5. As expected, we find no significant differential increase in coca crops in either coca suitable places or areas that expected to benefit from the crop substitution program.

In appendix A.6, we report and discuss a number of additional robustness tests, which account for serial correlation, use machine learning to select the set of controls, reestimate the baseline model without weighting by the size of municipalities, use a different measure of the predicted potential benefits from growing coca, include municipal-specific time trends, and compute p values using randomization inference.

VI. Mechanisms

We have documented a robust and large increase in the land cultivated with illegal coca crops in Colombia triggered by the anticipation effect that a preannounced policy generated among potential coca growers. This boost was concentrated in two types of areas in which the incentive provided by the announcement was expected to have a larger effect: areas with lower costs of growing coca given their soil and climate conditions and areas that anticipated a higher return from growing coca given by the estimated probability of benefiting from coca substitution incentives. We have also shown that the increase in coca growing is persistent over time, and it does not reverse with the actual implementation of the announced crop-substitution program.

We interpret these two pieces of evidence as consistent with an unintended negative consequence of a naive policy announcement, as well as with the endogenous effect that a surge in coca growing has on the dynamics of the internal armed conflict in Colombia. In particular, we hypothesize that the persistence of the effect is consistent with active armed groups attempting to appropriate the output of these crops in order to obtain resources from the drug trade. Because of this, once coca-growing areas were exposed to the presence of illegal groups, the effectiveness of the crop substitution campaign was reduced.

In this section, we provide further evidence in favor of our interpretation of the initial increase in coca growing as well as consistent with our explanation of its persistence. In particular, in the next subsection, we show that our findings

TABLE 3.—DRUG POLICY POTENTIAL EXPLANATIONS

Dependent variable: Coca production over 1,000 hectares	(1) Baseline	(2) Pre-2014 Eradication	(3) Exclude municipalities with previous eradication	(4) Distance to closest airport	(5) # Airports 80 radius	(6) Any airport 80 radious	(7) Coca seizures	(8) Cocaine lab destruction
Panel A: Suitability								
Suitability × Announcement	0.35***	0.34***	0.23**	0.35***	0.35***	0.34***	0.35***	0.31***
	(0.11)	(0.11)	(0.10)	(0.12)	(0.11)	(0.11)	(0.11)	(0.09)
R-squared	0.736	0.747	0.738	0.737	0.737	0.737	0.754	0.775
Panel B: Probability of crop substitution incentives								
Probability of substitution × Announcement	26.48***	33.41***	38.41***	26.53***	26.48***	26.45***	25.15***	23.09***
	(4.70)	(4.81)	(6.16)	(4.71)	(4.70)	(4.71)	(4.75)	(4.42)
Observations	8,736	8,736	7,736	8,736	8,736	8,736	8,736	8,736
R-squared	0.438	0.494	0.550	0.439	0.438	0.441	0.455	0.480
Municipalities	1,092	1,092	967	1,092	1,092	1,092	1,092	1,092
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean DV	0.274	0.274	0.128	0.274	0.274	0.274	0.274	0.274
SD DV	1.240	1.240	0.926	1.240	1.240	1.240	1.240	1.240

This table presents the results from the main specification in equation (1). *Suitability* is an index of coca suitability constructed by Mejía and Restrepo (2015). *Probability of substitution* is a predicted probability of the PNIS program estimated in table A2. *Announcement* is a dummy that takes the value 1 for the period after 2013. Column 2 adds the total area eradicated via aerial spraying between 2011 and 2014 over the municipality area interacted with year dummies. Column 3 excludes from the sample municipalities that experienced any aerial spraying between 2011 and 2014. Column 4 adds the log distance to the closest airport where the airplanes depart to conduct the aerial spraying interacted with year dummies. Column 5 adds the number of airports within a radius of eighty miles of the municipality centroid interacted with year dummies. Column 6 adds a dummy for a municipality having at least one airport within a radius of eighty miles of the municipality centroid interacted with year dummies. Column 7 adds the kilograms confiscated by seizures in every year at the municipality level for hydrochloride, coca paste, and coca leaves. Column 8 adds the number of cocaine clandestine laboratories destroyed in a given municipality in a given year. Panel A presents robust standard errors clustered at the municipality level in parentheses, and panel B presents bootstrap standard errors. * is significant at the 10% level, ** is significant at the 5% level, and *** is significant at the 1% level.

are most likely not driven by key alternative explanations. The second subsection formally explores the role of armed groups in the persistence of the upward shift in coca growing. The final subsection explores the state-monitoring capacity as a potential counteracting force of the perverse incentive given by the announcement.[26]

A. *Alternative Accounts*

Anti-drug policies. The period that followed the announcement coincides with anti-drug policy initiatives that took place in different parts of the country. In this subsection we corroborate that our results are not explained by timing and geographic reach of these policies, which include the ban of aerial coca spraying, drug seizures by the authorities, and the dismantling of clandestine cocaine-processing labs.

As discussed in section II, the banning of the aerial spraying of coca crops with the herbicide glyphosate is a key potential confounder. However, our findings are not consistent with this explanation. It is worth noting that the ban of aerial spraying became effective in mid-2015, more than a year after the joint announcement of the delegations of FARC and the Colombian government in Havana. As shown in figure 2, by then, coca growing was *already* differentially increasing in high suitability places and in areas with high predicted probability of receiving material incentives for crop substitution (panels C and D, respectively). The differential increase started in 2014, when coca growing in high suitability or predicted probability places departed from its 2011–2013 parallel trend.[27]

More formally, however, table 3 reports the results obtained from several statistical tests of the potential confounding effect of the change in the coca eradication policy. The baseline effects, previously reported in table 1, are reproduced for reference in column 1. In particular, panel A reports the coefficient associated with the interaction of the announcement time indicator with municipal-level coca suitability, and panel B reports the coefficient associated with the interaction of the same time indicator with the estimated probability of receiving the crop substitution incentives.[28] First, column 2 reports the coefficients obtained from estimating an equivalent specification, but adding as an additional control the cumulative share of the municipal area that was exposed to aerial eradication in 2011–2014, the period prior to its ban (see section II) interacted with the year fixed effects. By doing so, we flexibly control for differential trends at the municipality level parametrized by their exposure to aerial glyphosate spraying. The underlying assumption is that the aerial spraying of coca is highly persistent, and thus in the absence of the ban on aerial spraying, the fumigation of the crop would have

[26]Moreover, in appendix A.7 we explore whether the additional coca growing occurred in the vicinity of old coca crops or in new detached coca fields. The latter is more consistent with the announcement generating widespread incentives instead of private information for traditional farmers.

[27]The fact that the effect of the policy announcement on coca growing is contemporaneous is consistent with the observation that coca can be harvested as soon as six months after it is first planted. In Colombia it can be harvested four to six times a year. Comparatively, Peru and Bolivia see at the most four harvests per year (Walsh, Sánchez-Garzoli, & Salinas-Abdala, 2008).

[28]Specifically, we reproduce the estimates of columns 2 and 4 of table 1 (respectively in panel A and panel B of column 1 of table 3). The results are unchanged if we use as a baseline the most demanding specification of table 1, which adds as controls predetermined municipal characteristics interacted with the postannouncement time dummy.

taken place more or less in areas where it did occur prior to its ban.[29] Importantly, we show that the point estimate is largely unchanged in terms of both magnitude and significance. If anything, the differential effect increases slightly for the announcement on coca growing in municipalities expecting a large benefit from growing coca (column 2, panel B).

Second, in column 3 we estimate the main specification over the subsample of municipalities without any aerial spraying between 2011 and 2014. We thus show that our findings are still present in the subsample of coca-suitable areas (or areas with a high estimated probability of receiving crop substitution incentives) where coca was not eradicated prior to the announcement. Taking into account the standard deviation of the dependent variable in the estimation subsample, we find that the estimated coefficients are similar in magnitude to those of the full sample, and if anything they are larger for the case of the municipalities with a high predicted probability of benefiting from coca substitution incentives.

Third, we also take advantage of the production function of the aerial spraying campaign, specifically of the exact location of the airports from where eradication airplanes departed to flexibly control for the distance to each municipality. We also take into account that, for an airplane to conduct the aerial spraying and return to the airport safely given its fuel capacity, the maximum distance that it can cover from the departing airport is eighty miles.[30] Using these two pieces of information, we control for (i) the interaction between the log distance to the closest such airport with the year fixed effects (column 4 of table 3), (ii) the number of airports within a eighty-mile radius of the centroid of each municipality with the year fixed effects (column 5), and (iii) a dummy for having at least one airport within that radius interacted with the year fixed effects (column 6). In all these exercises, we find that the estimated coefficients associated with both of our treatments are remarkably similar to those in column 1 both in terms of magnitude and significance. These results support the previous finding, namely, that it is unlikely that our results are driven by the ban of aerial spraying in 2015.

In addition, it is important to show that the results are not driven by the effect of alternative anti-drug efforts. For instance, the ban of aerial spraying was accompanied by increased efforts by authorities to increase drug seizures. Drug cartels may have responded to this by increasing cultivation to compensate for the associated losses. We test this potential confounder in column 7 of table 3. We show that our baseline estimates are unchanged if we control for the amount (in kilograms) of coca products confiscated in interdiction operations by the authorities. Specifically, we add the time-varying seizures of coca leaves, coca paste, and hydrochloride in each municipality per year. Moreover, column 8 shows that controlling for the number of clandestine cocaine-producing labs (*cristalizaderos*) destroyed by the authorities at the municipality-per-year level does not change our estimates either.

FARC's favoring of their constituency. A second alternative account does recognize the unintended negative effect of the 2014 policy announcement but places the burden of responsibility on FARC. According to this interpretation (Lopez et al., 2019), following the announcement of the creation of PNIS, but before the signature of the peace agreement, FARC went back to their territory to promote coca plantations, ensuring the rural communities in their area of influence that benefits would come downstream if they did so.

Column 6 of table 3 adds as an additional control an indicator of FARC presence, as described in Prem et al. (2022, 2020), interacted with the time fixed effects. Again, this flexibly controls for differential trends at the municipality level parametrized by FARC presence prior to the announcement. Our coefficient of interest is slightly smaller (but the difference is not statistically significant) for the case of the interaction of the postannouncement indicator and the coca suitability index, and virtually unchanged for the case of the interaction of the time variation and the probability of receiving material incentives for crop substitution substitution. Therefore, our findings are not driven by this alternative interpretation.

Suitability of coca substitutes. Another potential alternative account has to do with the differential suitability of municipalities to grow (or exploit) potentially good substitutes for coca. Although it is worth noting that no legal crop offers anything close to the market value per unit of volume that coca does, some specific legal crops have been targeted by the Colombian government and by the international community as part of incentivized crop substitution schemes. Cacao and oil palm stand out because of their potentially large and steady international demand. On the one hand, Colombia is the largest oil palm producer in the Americas and one of the top five producers in the world. Oil palm has been at the core of Colombia's efforts of alternative development (i.e., crop substitution) since the mid-2000s with the support of the United States through USAID (Molano, 2008). On the other, since the start of the peace negotiations and in the face of an eventual peace agreement, the Colombian government and the agricultural producers' lobby group (Colombian Agricultural Society, SAC) have largely referred to cacao as the "peace crop," precisely because its attractiveness as a substitute for coca.[31] In addition, Colombia is one of the world's top coffee producers, and coffee is produced in about half of

[29] This assumption is reasonable. Different estimates have shown that coca eradication is very ineffective because of different strategies to protect coca bushes and the quick turnaround of the replanting efforts. In particular, Abadie et al. (2013) estimate that one acre of aerially eradicated coca decreases the areas covered by the crop by only 11% of an acre. Moreover, Mejía et al. (2015) estimate an effectiveness of the coca spraying campaign that is even smaller. This implies that, on average, one specific coca plantation has to be sprayed several times in order to be eradicated.

[30] Spraying airplanes are small aircraft and have to fly at relatively low altitudes.

[31] See *El cacao será el cultivo de la Paz*, available from https://sac.org.co/el-cacao-sera-el-cultivo-de-la-paz/ (last accessed 11/29/2022).

TABLE 4.—OTHER POTENTIAL ALTERNATIVE EXPLANATIONS

Dependent variable: Coca production over 1,000 hectares							
	(1) Baseline	(2) FARC presence	(3) Distance to ocean and Venezuela	(4) Suitability to coffee, palm, and cocoa	(5) Suitability to gold	(6) Social programs	(7) Previous deforestation
Panel A: Suitability							
Suitability × Announcement	0.35***	0.27***	0.34***	0.36***	0.35***	0.33***	0.26**
	(0.11)	(0.10)	(0.11)	(0.12)	(0.11)	(0.10)	(0.11)
R-squared	0.736	0.760	0.737	0.737	0.738	0.738	0.746
Panel B: Probability of crop substitution incentives							
Probability of substitution × Announcement	26.48***	25.47***	26.65***	26.48***	26.47***	26.38***	26.03***
	(4.70)	(4.65)	(4.73)	(4.71)	(4.73)	(4.66)	(4.67)
Observations	8,736	8,736	8,736	8,736	8,736	8,736	8,736
R-squared	0.438	0.472	0.439	0.438	0.443	0.442	0.454
Municipalities	1,092	1,092	1,092	1,092	1,092	1,092	1,092
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean DV	0.274	0.274	0.274	0.274	0.274	0.274	0.274
SD DV	1.240	1.240	1.240	1.240	1.240	1.240	1.240

This table presents the results from the main specification in equation (1). *Suitability* is an index of coca suitability constructed by Mejía and Restrepo (2015). *Probability of substitution* is a predicted probability of receiving crop substitution incentives estimated in table A2. *Announcement* is a dummy that takes the value 1 for the period after 2013. Column 2 adds a dummy for a municipality having at least one FARC attack between 2011 and 2014 interacted with year dummies. Column 3 adds the log distance to the ocean and the frontier with Venezuela. Column 4 adds coffee intensity from the 2005 census and the share of the municipality with high suitability to cocoa and palm. Column 5 adds a dummy that takes the value 1 if there was gold production in the municipality in 2011 as well as a suitability for gold measure based on Idrobo et al. (2014). Column 6 adds as a control the families per capita that are beneficiaries of Programas Familias en su Tierra and IRACA. Column 7 adds the cumulative deforestation between 2005 and 2013 over the municipality area interacted with year fixed effects. Panel A presents robust standard errors clustered at the municipality level in parentheses, and panel B presents bootstrap standard errors. * is significant at the 10% level, ** is significant at the 5% level, and *** is significant at the 1% level.

the Colombian municipalities, offering a stable source of income to hundreds of thousands of rural households. Finally, legal and illegal gold mining may generate rents and increase the opportunity cost of growing coca (especially during our sample period, when the international price of gold was at its pre-2005 levels).

Importantly, when we interact the time fixed effects with the suitability of every municipality to grow oil palm and the suitability to grow cacao (both obtained from the Unidad de Planificación Rural Agropecuaria, UPRA), as well as with the share of the municipal area that produced coffee in 1997 (obtained from Dube & Vargas, 2013), our baseline estimates of the main effects are unchanged (see column 4 of table 4). The same is true if we control for the interaction of the year fixed effects with a measure of geochemical anomalies of the municipal soil associated with the presence of gold deposits (obtained from Idrobo, Mejía, & Tribin, 2014, see column 5).[32]

However, to the extent that these alternative economic activities do play some role in allowing coca farmers to substitute their plantation with legal crops or mining, we should expect that the effect of the announcement about coca growing in coca-suitable places and in areas with a high estimated probability of benefiting from substitution incentives would be attenuated. In columns 5 to 12 of table A7 we find evidence consistent with this, at least for the case of coca suitability.

Three additional alternatives, namely, municipalities with differential cost advantage to trafficking cocaine, the contemporaneous roll-out of social programs, and the endogeneity of coca suitability to past deforestation, are discussed (and falsified) in appendix A.8.

B. *Persistence*

Panels C and D of figure 2 suggest that the differential postannouncement increase in coca growing in coca-suitable municipalities and in areas that expected to benefit from the crop substitution programs persisted even after the actual implementation of the announced crop-substitution incentives program. In this subsection we examine two potential reasons for this persistence.

The role of other armed groups. We present two different pieces of evidence that suggest that the persistence in the surge of coca cultivation over time after the announcement was driven by its effect on the dynamics of the internal armed conflict in Colombia. First, in table A7 we estimate heterogeneous effects parametrized with an indicator of the presence of (or in the vicinity to) at least one illegal armed group in municipality *m* before the policy announcement.[33] We find a positive coefficient in the triple-interaction term, suggesting that in places with low cost for growing coca (or with high

[32] A less flexible specification that interacts with gold anomalies measure with the international price of gold instead of with the year fixed effects also yields unchanged estimates of the effects of interest.

[33] Measuring the influence exercised by an armed group over a specific location is extremely challenging. Indicators of presence and nonviolent coercion over a large set of municipalities cannot be systematically recorded in an objective way. Violence, on the other hand, although more easily observed, is only imperfectly correlated with territorial dominance. However, nonviolent dominance is unlikely to occur without any violence inflicted in the past. It is thus reasonable to assume that the ability to inflict localized violence over a certain period could be expected to translate into influence in different ways. We thus follow a growing empirical literature on the Colombian conflict (see, e.g., Acemoglu, Robinson, & Santos, 2013 and Ch et al., 2018) and use past violence over a period of years as an (imperfect) indicator of influence, as in Prem, Rivera et al. (2022).

potential benefits of the substitution program), which also had the presence of armed groups before the policy announcement, the increase in coca was differentially larger after the policy announcement. Specifically, in column 1 of table A7 the triple interaction term doubles the magnitude of interaction term for coca suitability, whereas in column 3 it increases the average effect of the probability of benefiting from coca substitution incentives by around 160%. These results suggest that the increase in coca growing in areas attractive to armed groups was not only larger, but also economically significant. This is consistent with the persistence of the coca boom being, at least partially, driven by the endogenous dynamics of the conflict, whereby existing armed groups took advantage of the newly available coca rents to dominate the hosting territories.

To flesh out this hypothesized channel, and because territorial dispute is often violent in the context of an internal conflict—which has been the case of Colombia for more than five decades—we complement this evidence by exploring how armed conflict changed between 2013 (the year before the policy announcement) and 2018 for municipalities with high suitability or high probability of receiving material incentives for crop substitution and municipalities with low suitability or low probability. For this purpose, we use our digitized armed conflict data discussed in section III.

Specifically, for each municipality we took the difference between a dummy that takes the value 1 if the municipality faced an attack by an armed group other than FARC in 2018 minus the same dummy computed in 2013.[34] Thus this variable takes a value of 1 if in 2018 there was at least one attack and there was no attack before, a value 0 if the municipality had an attack in both 2018 and 2013 or did not face any attack in neither year, and a value of -1 if the municipality had an attack in 2013 but not in 2018. We split municipalities into high and low suitability and high and low probability using the median of the empirical distribution of these measures. It can be seen from all panels in figure A4 that municipalities with low suitability and low probability faced a reduction in the probability of an attack by other armed groups, whereas municipalities with high suitability or high probability faced an increase in the probability of an attack, with the increase almost doubling the reduction. Moreover, the difference between the averages presented in the bars is statistically significant with p values smaller than 3% in all cases.

Table A8 shows a similar pattern in a regression framework that estimate the cross-sectional correlation between a dummy that takes a value of 1 if a municipality experienced at least one violent event (panel A) or at least one attack (panel B) perpetrated by a non-FARC armed group in 2018 (dependent variable) and our two proxies of the differential incentive provided by the policy announcement, which account for the cost advantage of growing coca and for the expected benefits of doing so. Importantly, in addition to municipal characteristics we control for both department-level fixed effects, a set of municipality characteristics measured before 2013, and for the lagged dependent variable, as measured in 2013 (the year prior to the announcement). Looking at our preferred specification (even columns), the results suggest that a one-standard-deviation increase in the suitability index is associated with a probability of experiencing a violent event by one percentage point (panel A, column 2), which corresponds to an increase of 18% with respect to the average of the dependent variable. In the case of the estimated probability of benefiting from crop substitution incentives, an increase of 10% in this probability relative to the mean is associated with an increase in 0.3 percentage points in the probability of experiencing a violent event in 2018 (panel A, column 4), which corresponds to an increase of 5% with respect to the average of the dependent variable. If we look at unilateral attacks instead of any type of violent events (panel B) the results are virtually unchanged.

Postimplementation material incentives. The PNIS program was created in May 2017, six months after the peace agreement was ratified by the Colombian Congress. As of the end of our sample period (2018) it had reached almost 99,000 farmers in 56 municipalities, and two-thirds of them had received payments (Garzón et al., 2019). The partial adoption of PNIS allows us to test if the program implementation actually incentivized coca growing in these areas above and beyond the incentives provided by coca suitability and the predicted probability of receiving the incentives for crop substitution. To be sure, to qualify for the subsidies, targeted farmers cannot return to growing coca after the crop substitution process has started. However, other (nontargeted) farmers within the same municipality may be encouraged to grow coca if they learn that other farmers who did so are getting material incentives. We thus estimate our baseline specification (1) but add the interaction term between an indicator of whether at least one farmer had been selected by PNIS in a municipality and a time dummy that equals one in 2018.

Consistent with our incentives-based story, in table A9 of the appendix, we find that not only did the announcement of future benefits foster coca growing in municipalities with a high cost advantage or expected benefits, but also that these perverse incentives were present in the actual implementation of the program.[35]

[34]We exclude FARC because they signed a peace agreement with the government in 2016, so they are excluded from the original data set as perpetrators of conflict-related events.

[35]Moreover, the material incentives associated with participation in PNIS may have encouraged farmers who had been formerly dispossessed of their land to try to get it back from the Land Restitution Unit that was created in 2012. We thus test the correlation between the cumulative number of land restitution requests and our measures of coca suitability and predicted probability of benefiting from crop substitution incentives. These results are presented in table A10 of the appendix. Although only suggestive because of their cross-sectional nature (the publicly available data on land restitution claims provide only municipal totals since the start of the Land Restitution policy in 2012 and not a yearly panel), when controlling for department fixed effects and large battery of municipality controls, we find a positive association between our measures of coca-growing cost advantage

C. *State Presence and Monitoring*

The final analysis that we undertake is to explore the extent to which state presence and monitoring at the local level helps attenuate the negative unintended consequences of the policy announcement. In particular, we first explore potential heterogeneous effects in terms of a proxy of state presence. Specifically, we take the total per capita number of public offices in each municipality and standardize it to facilitate the interpretation of the heterogeneous effects test. The results are reported in columns 3 and 4 of table A7. We find that the increase in coca growing after the announcement in both coca suitable places and in areas with a high predicted probability of receiving material incentives for crop substitution is attenuated in places with stronger state capacity. Although the effect is economically large in both cases, it is significant only at conventional levels for the specification that includes coca suitability (column 3).

Second, we follow Bonilla-Mejía and Higuera-Mendieta (2019) and Prem et al. (2020) and use protected areas within municipalities, that is, places with relatively higher monitoring capacity, to explore further potential heterogeneous effects. Protected areas include national parks with native and sometimes endangered species, indigenous lands, and communal lands. There, formal or informal (or else traditional ethnic-like) institutions are more likely to be present and active. Moreover, they are additionally monitored by park rangers who report any illegal activity that takes place within their boundaries to the central government. The results are reported in table A11, which presents a version of equation (1) that uses as dependent variable the share of the municipality with coca cultivation that is in a protected area (columns 1 and 3) and that it is not in a protected area (columns 2 and 4). We find that in nonprotected areas an increase of one standard deviation in the suitability index increases the share of coca cultivation by 0.29 per 1,000 hectares. This increase represents a 152% increase with respect to the average share of coca before the announcement (column 2). We also find a statistically significant increase in protected areas, with an increase of 72% with respect to the mean before the announcement (column 1). The difference between both effects is statistically significant (p value of 0.02). We find similar patterns in the case of the potential benefits associated with the substitution program (see columns 3 and 4).[36]

These results suggest that, within municipalities, areas with more monitoring capacity, experienced a smaller increase in the area cultivated with coca after the appearance of the incentives generated by the policy announcement. Also, the fact that we observe an increase in protected areas is additional evidence that our results are not driven by the 2015 ban in aerial spraying, because aerial eradication was not allowed in those areas before the ban.

and expected benefits and several outcomes associated with land restitution requests.

[36]The attenuation effect of protected areas on the postannouncement increase in coca growing can be graphically seen in figure A6 of the appendix.

VII. Conclusion

This paper contributes to the policy discussion of what are the most effective strategies to win the war on drugs. It does do by discussing how ill-designed incentive-based policies can be ineffective and have unintended negative consequences, as has been widely shown for prohibition and enforcement-type policies in a variety of contexts. Specifically, we exploit the recent experience of Colombia, which witnessed an increase in coca growing from 48,000 hectares in 2013 (the lowest figure since 1999) to 171,000 hectares in 2017 (the historical record). We document that this large increase was to a great extent the result on a promise made in May 2014 that material benefits would be given to coca growers in the future. Importantly, we show that our results are not driven by alternative explanations such as the ban of aerial spraying in 2015, an explanation recently championed for political reasons by some political parties in Colombia (as well as by the U.S. government).

Indeed, the increase in coca cultivation over the past five years has generated an intense technical and political debate about its causes and consequences. Our work contributes to this debate by showing how a naive and untimely policy announcement during the peace negotiations between the national government and the FARC generated perverse incentives that led to an unprecedented historical increase in coca cultivation in the country. Our most demanding specifications show that the announcement of material benefits for coca growers who decided to embark on substitution programs increased coca cultivation in some areas of the country that are suitable for coca cultivation by more than 100%.

Our results highlight the importance of conducting well-designed policy announcements that could minimize unintended consequences derived from the policy-generated incentives. Specifically, the design of illicit crop substitution programs should incorporate baseline estimations of illicit crop cultivation at very desegregated levels and should impose conditions for the delivery of material benefits for growers who decide to substitute their illicit crops. Otherwise, the announcement of benefits may cause the opposite effect to that initially expected.

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